Irfan Mas'udi

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Summary

AI/ML Engineer with 5+ years bridging research and production. Currently at Samsung Research Indonesia, contributing to recommendation systems, LLM-based agentic workflows, and generative AI for consumer devices. Known for translating complex research ideation into production-ready solutions.

Professional Experience

Samsung Research Indonesia, Jakarta, Indonesia — Feb 2024 – Present
R&D Institute of Samsung Electronics Co., Ltd
AI/ML Engineer

- Deployed large-scale RecSys for Samsung Galaxy Store (Apps, Games, Themes, Wallpapers); applied content-based and collaborative filtering leveraging multi-modal representation learning and multi-stage approach to generate top-K personalized recommendations, reducing inference latency by ∼90% and improving CTR baseline by ∼+30 p.p.
- Designed Deep Learning based GamingHub RecSys that utilize historical recency user interaction with category-aware (e.g., cloud game and instance play) which surpassed baseline metrics (nDCG, F1, Coverage, Diversity).
- Designed a parallel multi-LLM orchestration system that processed requests 3–5× faster across billions of items.
- Deployed an LLM-based RecSys through context engineering, reducing hallucinations, cut token usage by 70–90%, and costs by 10–30%.
- Developed a cross-store app matching system that uses ML-based feature engineering and designed distributed inference pipelines for multimodal LLM, making processing ∼80× faster than a standard local-serving approach and reducing API costs by 80–93%.
- Led a team of 5 engineers to research LLM-based agentic AI workflows — including Retrieval-Augmented Generation (RAG), prompt engineering, structured outputs, function calling, and tool integration as a context engineering — for Samsung Vision AI Companion feature.
- Led a team of 3 to research deep generative models for image and 3D applications, enabling immersive AI features on Samsung devices.
- Built and scaled end-to-end MLOps pipelines — data ingestion, feature engineering, model training, evaluation, CI/CD, and deployment — across cloud and on-device environments.
- Contributed to multiple AI patents and peer-reviewed publications; awarded Employee of the Quarter Q2 2025.
- **Tech:** Python, PyTorch, TensorFlow, Databricks, AWS, Apache Spark, SQL, MLflow, Docker, FAISS, vLLM, Vector Database, LangGraph, LlamaIndex, Pandas, Polars, Git.

EEPIS Robotics Research Center, Surabaya, Indonesia — Sep 2018 – Nov 2019
University research center for robotics technology and development
Research Assistant

- Commercialized a research-grade hybrid PID–fuzzy control system for ROV motion, improving maneuverability and stability by 21.9% over a conventional PID baseline.
- Applied YOLOv3 to build an underwater object classification pipeline achieving 89.6% detection accuracy; led full cycle from dataset collection and annotation to model training and evaluation.
- First-authored and co-authored multiple research papers published in international conferences and journals.
- **Tech:** Python, TensorFlow, ROS, OpenCV, Git.

Advanced Robotics Technology, Surabaya, Indonesia — Mar 2017 – Sep 2018
Pre-seed technology startup specializing in educational robotics manufacturing
Software Engineer

- Led research-to-product commercialization of educational robotics systems, directing 4 engineers across embedded (C/C++) and Android application development.
- Co-led fundraising efforts securing IDR 120M in pre-seed investment; built data-driven tools for production optimization including inventory tracking and sales analytics.
- **Tech:** C, C++, Java (Android), Excel.

Technical Skills

- **Languages:** Python, SQL, C, C++, Java
- **ML/DL Frameworks:** PyTorch, TensorFlow, Scikit-Learn, HuggingFace, LlamaIndex
- **Generative AI & LLM:** Diffusion models, GANs, LLM fine-tuning, RAG, prompt engineering, agentic workflows, structured outputs, function calling, MCP, model harness, and Context Engineering.
- **Recommender Systems:** Collaborative filtering, content-based filtering, matrix factorization (SVD, ALS, BPR), two-tower models, LightGCN, FAISS, Vector Search & Databases.
- **MLOps & Infrastructure:** MLflow, Docker, AWS, Databricks, Apache Spark, Apache Kafka, CI/CD, Git
- **Data Engineering:** Pandas, Polars, ETL pipelines, NoSQL (MongoDB, Redis, Cassandra), PostgreSQL, MySQL
- **Other:** A/B testing, time series analysis (ARIMA), reinforcement learning, computer vision (OpenCV, YOLO, CNN-based, U-Net, Transformer-based, DINO)

Selected Projects [More]

- **LLM from Scratch:** Implemented BPE tokenizer, Transformer decoder architecture, and autoregressive pretraining pipeline end-to-end in PyTorch.
- **Vision Transformer from Scratch:** Implemented ViT architecture and training pipeline from scratch in PyTorch for image classification.
- **Semantic Segmentation:** Built U-Net for urban scene understanding; achieved 88.5% test accuracy and 0.40 mean IoU.
- **GAN for Image Generation:** Implemented Generator and Discriminator from scratch for MNIST image synthesis.
- **Human Activity Recognition:** Benchmarked LSTM, BiLSTM, and 1D-CNN on multivariate time series; best F1 of 0.630 with 1D-CNN.
- **Multivariate Time Series Forecasting:** Two-layer BiLSTM on 7-feature sequential dataset; RMSE of 4.03.
- **Clustering Benchmarking:** Compared LDA, QDA, KNN, and SVM; best F1 0.800 with SVM. K-Means clustering with PCA on MNIST achieving 0.790 purity.

Education

Politecnico di Milano, Milano, Italy — Feb 2020 – Dec 2023
MSc in Computer Science and Engineering — GPA: 3.45/4.0 (86/110)

Key coursework: Machine Learning, Deep Learning & ANNs, Recommender Systems, Unstructured and Streaming Data Engineering, Advanced Databases, Operation Research, Time Series Analysis, Reinforcement Learning.

Thesis: DILLEMA: Metamorphic Testing for Deep Learning using Diffusion and Large Language Models.

Politeknik Elektronika Negeri Surabaya, Surabaya, Indonesia — Jun 2014 – Aug 2018
BASc in Computer Engineering — GPA: 3.67/4.0

Key coursework: Programming, Algorithm and Data Structure, Object-Oriented Programming, Computer Vision.

Selected Publications [More]

DILLEMA: Diffusion and Large Language Models for Multi-Modal Augmentation
L Baresi, DYX Hu, MI Mas'udi, G Quattrocchi
IEEE/ACM International Workshop on Deep Learning for Testing (DeepTest), 2025

EROV: Depth and Balance Control for ROV Motion Using Fuzzy PID Method
EH Binugroho, W Ab, MI Mas'udi et al.
International Electronics Symposium (IES), 2019

Fusion of Feedforward and Feedback Control Using Fuzzy for Active Handling and Dribbling in MSL Robot Soccer
MI Mas'udi, F Ardilla, IK Wibowo
International Electronics Symposium on Engineering Technology and Applications (IES-ETA), 2018

Awards & Recognition

- Employee of the Quarter, Samsung Research Indonesia — 2025
- Invited guest lecturer on AI/ML topics at universities — Present
- LPDP Scholarship (Indonesian Government fully-funded Master's scholarship) — 2019
- Runner-up, Most Outstanding Student of the Republic of Indonesia — 2017
- 2nd Runner-up & NOK Award, Asia-Pacific Broadcasting Union (ABU) ROBOCON — 2016
- Winner, Indonesia National Robot Contest — 2016